

The Krabby Real Estate Formula

You can download the initial exercise file from [link](#).

Welcome to "The Krabby Real Estate"! Mr. Krabs, the famous restaurant owner, has recently realized that the Krabby Patty formula is no longer the only way to get rich in the city of "Bikini Bottom". He has figured out that the real money lies in the volatile real estate market!

But a massive crisis has struck the company! Plankton, using his computer wife's AI, is pre-purchasing the city's best houses with highly precise and engineered prices. Meanwhile, Mr. Krabs discovered that Squidward (who until yesterday was responsible for pricing and appraising the houses) was setting the prices entirely by chance, based on his mood, and just by looking at the exterior of the houses! Mr. Krabs fired Squidward out of pure anger and now senses the danger of his new empire going bankrupt. Now, Mr. Krabs wants one thing from you: build a system that predicts the exact price of houses in Bikini Bottom before Plankton gets his hands on them.

SpongeBob's team (who now collects data across the city instead of flipping burgers) has extracted a dataset of sold houses in various neighborhoods and provided it to you. Your task is to design this model.

The data collection team has provided the extracted information in the form of two files (train and test). Each row of this dataset represents the specifications of a house. The details of the columns in this dataset are as follows:

Column Name	Description
lat	House latitude (map coordinates)
long	House longitude (map coordinates)

Column Name	Description
area	House area (in square meters)
age	Building age (years)
total_floors	Total number of building floors
floor_number	Floor number where the unit is located
rooms	Number of bedrooms
document_type	Property deed type (Private, Endowment, Unregistered Deed, etc.)
cooling_type	Cooling system (Split, Chiller, Water Cooler, etc.)
monthly_levy	Monthly building maintenance fee
has_master_bedroom	Does the house have a master bedroom? (1: Yes, 0: No)
parking_type	Parking status and type (Covered, Courtyard, etc.)
neighbor_noise_level	Neighbors' noise level (Low, Average, High)
distance_to_hospital_km	Distance to the nearest hospital (kilometers)
has_smart_home	Equipped with a smart home system (1: Yes, 0: No)
water_pressure	Building water pressure status
exterior_style	Building exterior style (Roman, Modern, Classic)
fiber_internet	Access to fiber optic internet (1: Yes, 0: No)
manager_present	Presence of a building manager or lobbyman (1: Yes, 0: No)
window_type	Type of windows (Double-glazed, Single-glazed, etc.)
price	Final sale price of the house

Output

To evaluate your program, you must predict the prices of the houses in `test.csv`. Your output should be a text or CSV file containing only **one column** named `price`. Each row in this column must exactly represent the predicted price for the corresponding row in the input data.

Code snippet

```
price
2450000
1860500
3400000
985000
...
```

Evaluation Method

The following formula is used to evaluate your system:

$$Score = \max \left(0, 100 \times \left(1 - \frac{1 - R^2 Score}{0.04} \right) \right)$$